100-Species Animal Classifier: Fine-Tuned CNN for Indian Wildlife & Beyond

Course: Statistical Methods in AI (SMAI) — Assignment 3, Tier 2 **Theme:** T7.7 — 100-Species Fine-Tuned Identifier **Dataset:** ViratGarg/animal_species_SMAI

Team Name: Kuch bhi daal do

Name	Roll No.	Email
Krrish Goenka	2023112023	krrish.goenka@research.iiit.ac.in
Krish Agarwal	2023113017	krish.agarwal@research.iiit.ac.in
Pranav Shankar	2023112011	pranav.shankar@research.iiit.ac.in
Virat Garg	2023101081	virat.garg@students.iiit.ac.in
Manas Agrawal	2023113023	manas.agrawal@research.iiit.ac.in

Abstract

We present a multi-approach animal species classification system unifying three wildlife datasets — mammals, birds, and butterflies — into a single 100-class benchmark. Three strategies are evaluated: (1) a ResNet50 fine-tuned end-to-end on all 100 classes, (2) a two-stage hierarchical ResNet50 pipeline that routes images by animal group before species identification, and (3) a CLIP-based approach using KNN and SVM over frozen visual embeddings. All approaches exceed 95% test accuracy, with fine-tuned ResNet pipelines reaching up to **97.22%**.

---

1. Dataset

The dataset merges three animal image sources into a unified 100-class label space, hosted on HuggingFace:

Subset	Species	Images
Mammals	45	13,751
Birds	25	8,750
Butterflies	30	4,158
Total	**100**	**26,659**

Splits: Train: 21,327 | Val: 2,666 | Test: 2,666

---

2. Methods

2.1 Model 1: Single ResNet50 (Direct 100-Class Fine-Tuning)

A ResNet50 pretrained on ImageNet1K is used as the backbone. The final FC layer is replaced with a sequential head consisting of a Dropout layer (p=0.4) followed by a linear layer mapping to 100 output classes. The entire network is fine-tuned end-to-end.

Hyperparameters:

Parameter	Value
Optimizer	AdamW
Learning Rate	1e-4
Weight Decay	1e-4
Epochs	20
LR Schedule	Cosine Annealing (T_max=20)
Loss	Cross-Entropy (label smoothing ε=0.1)
Dropout	0.4
Batch Size	32
Input Size	224×224

2.2 Model 2: Two-Stage Hierarchical ResNet50 Pipeline

Four ResNet50 models are used in a cascaded pipeline:

```
Input Image
     │
     ▼
┌─────────────────────────────────────────┐
│  Stage 1: Group Classifier (3-way)  │
│      Bird | Butterfly | Mammal       │
└─────────────────────────────────────────┘
        │            │            │
        ▼            ▼            ▼
    Birds       Butterfly    Mammals
    ResNet50    ResNet50     ResNet50
    (25-class)  (30-class)   (45-class)
```

Stage 1 classifies images into one of three animal groups; the predicted group routes each image to the corresponding Stage 2 species model. Each Stage 2 model tackles at most a 45-way classification, rather than the full 100-way problem.

Hyperparameters (same across all 4 models):

Parameter	Value

Parameter	Value
Optimizer	AdamW
Learning Rate	1e-4
Weight Decay	1e-4
Stage 1 Epochs	15
Stage 2 Epochs	15
LR Schedule	Cosine Annealing
Loss	Cross-Entropy (label smoothing ε=0.1)
Dropout	0.4
Batch Size	32

Checkpointing is done on best validation accuracy for each model.

2.3 Model 3: CLIP Embeddings + KNN / SVM

No fine-tuning is required. Frozen image encoders extract visual embeddings which are classified using classical ML methods.

Encoder	Description
OpenAI CLIP ViT-B/16	General-purpose, trained on 400M image-text pairs
OpenAI CLIP ViT-B/32	Lighter general-purpose variant
BioCLIP	Domain-specialized CLIP for biological/taxonomic data

Classification heads: Zero-Shot (cosine similarity to text prompts), KNN (k=1 to 20), Linear SVM.

---

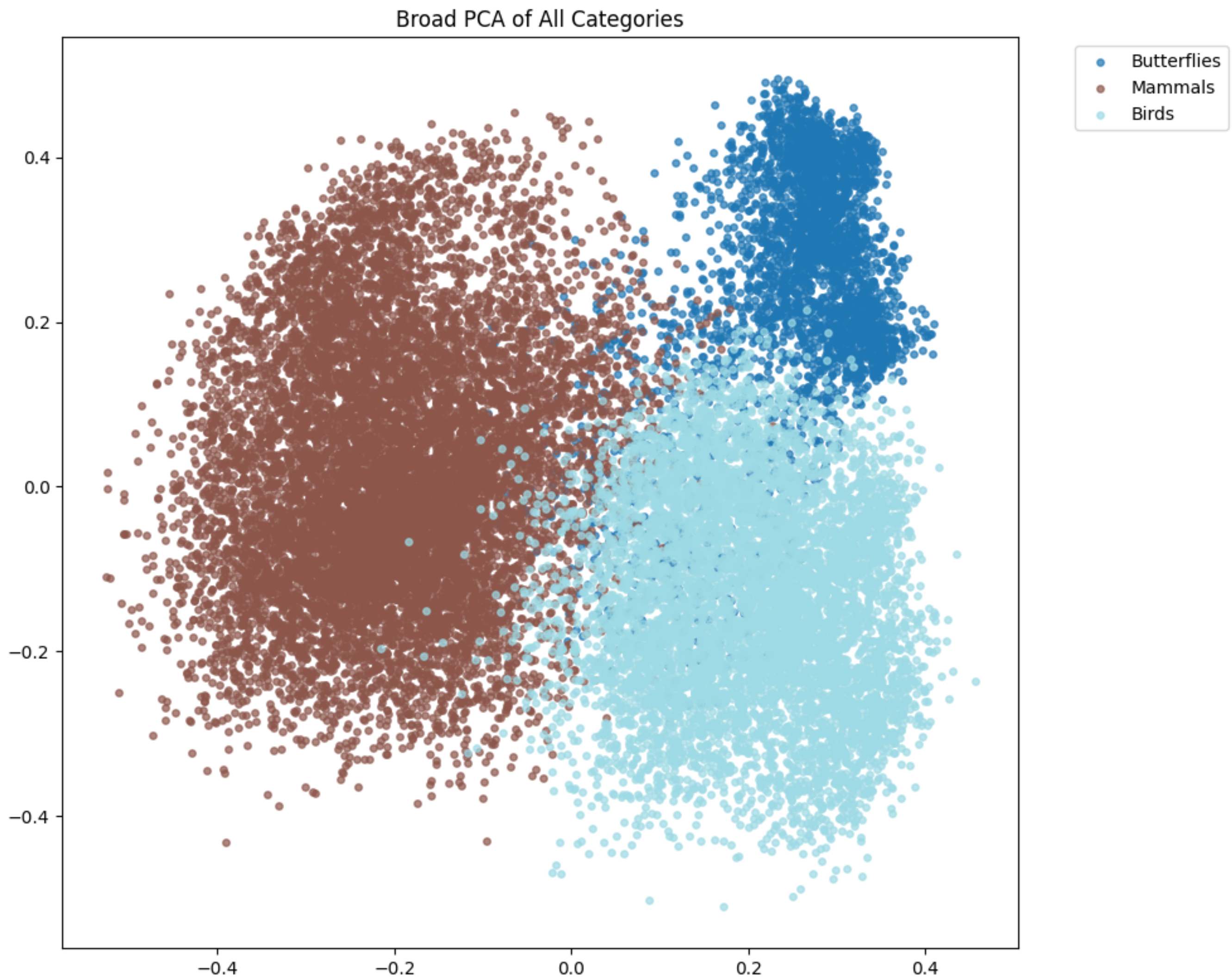
3. Feature Space Analysis

PCA projections of CLIP (ViT-B/16) embeddings reveal macro-level separation across the three animal groups — butterflies form a tight, well-separated cluster while mammals and birds overlap moderately at boundaries. This confirms CLIP encodes high-level biology without any task-specific training.

Silhouette Scores:

Scope	Score
3 broad categories	0.1449
Mammals (45 species)	0.1165
Birds (25 species)	0.1192
Butterflies (30 species)	0.0600

The low butterfly silhouette score (0.06) foreshadows weaker KNN/SVM performance on that subset — butterfly species share considerable visual overlap in the embedding space.



---

4. Results

4.1 Model 1: Single ResNet50

Epoch	Train Acc	Val Acc	Test Acc
1	78.09%	94.37%	94.11%
5	98.18%	96.06%	95.87%
10	99.34%	96.70%	96.51%
15	99.84%	97.00%	97.11%
20	99.96%	97.41%	**97.22%**

Final Test Accuracy: 97.22% | Macro Specificity: 99.97%

The model converges rapidly — validation accuracy exceeds 94% by epoch 1. A small but stable generalization gap persists (~2.5%), controlled by label smoothing and cosine annealing.

4.2 Model 2: Two-Stage Hierarchical Pipeline

Stage 1 — Group Classifier:

Metric	Score
Accuracy	99.89%
Macro F1	99.86%
Macro Specificity	99.93%

Only 3 out of 2,666 test samples were misrouted (0.11%). Misrouted samples cannot be recovered regardless of Stage 2 performance.

Stage 2 — Species Classifiers:

Group	Test Accuracy	Macro F1
Birds (25 classes)	97.14%	97.13%
Butterflies (30 classes)	97.59%	97.54%
Mammals (45 classes)	96.66%	96.59%

End-to-End Pipeline:

Metric	Score
End-to-end Species Accuracy	**96.89%**
End-to-end Macro F1	96.91%
End-to-end Macro Specificity	99.97%

4.3 Model 3: CLIP Embeddings

SVM Results:

Embeddings	Mammals	Birds	Butterflies	Combined
CLIP ViT-B/16	95.86%	97.54%	91.71%	95.76%
BioCLIP	94.62%	**99.03%**	**98.08%**	**96.61%**

BioCLIP dramatically improves butterfly accuracy (91.71% → 98.08%), as expected given its biological taxonomy training data.

Zero-Shot Results (no training data used):

Model	Mammals	Birds	Butterflies	Combined
CLIP ViT-B/32	92.26%	79.09%	37.86%	78.72%

Model	Mammals	Birds	Butterflies	Combined
CLIP ViT-B/16	93.82%	83.49%	39.42%	81.14%
BioCLIP	81.46%	89.71%	55.17%	80.01%

Zero-shot butterfly accuracy is low for standard CLIP models, confirming that the bottleneck is text-prompt matching rather than the image encoder quality.

4.4 Summary Comparison

Approach	Test Accuracy
Model 1: Single ResNet50	**97.22%**
Model 2: Hierarchical ResNet50	96.89%
Model 3: BioCLIP + SVM	96.61%
Model 3: CLIP ViT-B/16 + SVM	95.76%
Model 3: CLIP + KNN (best k)	~94–95%
Zero-Shot CLIP ViT-B/16	81.14%
Zero-Shot BioCLIP	80.01%

---

5. Demo Application

We built a Streamlit web application that allows users to upload an image and get species predictions from all three models. The app returns the top-3 predicted species along with confidence scores, and displays the model's routing decision in the hierarchical pipeline.

> **Screenshot placeholder — insert app screenshot here:**

Streamlit App Screenshot

The app takes an image as input, runs inference through the selected model, and returns the top predicted species with confidence scores.

---

6. Discussion

Why are accuracies so high? ResNet50 pretrained on ImageNet1K and CLIP trained on 400M image-text pairs are both significantly overqualified for 100-class animal classification. These models already encode deep features for fur texture, wing patterns, and body shape — fine-tuning only needs to align the final classification head to the new label space. Additionally, most species in our dataset are visually distinctive — a giraffe looks nothing like a warthog — unlike fine-grained benchmarks such as CUB-200 where inter-class differences can be subtle.

Why does CLIP underperform fine-tuned ResNets? KNN and SVM find proximity/linear boundaries in a 512-dimensional space not explicitly trained for species-level separability. Fine-tuned ResNets adapt the entire representation to maximize species-level discrimination. That said, the use of KNN and SVM here was a deliberate pedagogical choice — integrating classical ML algorithms from the course curriculum with modern CLIP embeddings demonstrates that strong representations can unlock solid performance even from simple classifiers.

Butterflies are the hardest class. The lowest silhouette score (0.06), lowest KNN accuracy (~88%), and worst zero-shot scores all point to butterflies as the most challenging group. BioCLIP's dramatic improvement on butterflies confirms this is a representation issue rather than an inherent difficulty — the right pretraining data matters.

---

7. Limitations

- **Butterfly discrimination** remains the weakest link across all methods, particularly for zero-shot CLIP.
- **Seal and blue whale** show the lowest per-class F1 scores (seal: 0.828, blue whale: 0.840) among mammals, likely due to visual similarity with sea lions and lack of close-up distinguishing features respectively.
- **Small test set per class** (~26 samples average) means per-class metrics have high variance.

---

8. Conclusion

Three approaches to 100-class animal species classification all achieve >95% test accuracy on a combined mammals/birds/butterflies benchmark. The single ResNet50 direct fine-tune achieves the best overall accuracy (97.22%), the hierarchical pipeline (96.89%) offers better modularity and interpretability, and the CLIP+SVM approach (up to 96.61% with BioCLIP) requires no gradient-based training on the target dataset. High accuracies across the board reflect the strength of modern pretrained representations applied to a visually distinctive, well-curated dataset.

References

1. He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. *CVPR*.
2. Radford, A., et al. (2021). Learning Transferable Visual Models From Natural Language Supervision (CLIP). *ICML*.
3. Stevens, S., et al. (2024). BioCLIP: A Vision Foundation Model for the Tree of Life. *CVPR*.
4. Cover, T. M., & Hart, P. E. (1967). Nearest Neighbor Pattern Classification. *IEEE Transactions on Information Theory*.
5. Cortes, C., & Vapnik, V. (1995). Support-Vector Networks. *Machine Learning*.
6. Dataset: ViratGarg/animal_species_SMAI on HuggingFace.

Acknowledgements

LLM tools were used for code scaffolding assistance. All model training, evaluation, analysis, and report writing were carried out by the project team.

SMAI Assignment 3 — Tier 2 | T7.7: 100-Species Fine-Tuned Identifier